Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State what is meant by a member in tension.

- The member is being pulled / stretched by the forces at its ends

3 Quick check

- a) CWM = ACWM
- b) 100 N m
- c) Strut
- d) 980 N

4 Key terms

- 1 – C
- 2 – E
- 3 – B
- 4 – D
- 5 – A

5 State the two conditions of equilibrium.

- The sum of the forces = 0 (upward = downward, left = right)
- The sum of the moments = 0 (clockwise = anticlockwise)

6 Explain what is meant by ‘Free body diagram’.

- One object shown with all the forces acting on it.

7 Explain what is meant by ‘Concurrent forces’.

- Forces acting through a single point.